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1.9 Pre-Quiz: Arithmetic and Geometric Sequences

Review for Wednesday's quiz. A calculator is allowed. Show your working: write the formula you use, then substitute.

1. Consider the arithmetic sequence

12, 19, 26, 33, ...

- (a) Write down the value of the common difference d .
- (b) Find an expression for u_n , the n^{th} term.
- (c) Find u_{40} .
- (d) Show that 250 is a term of the sequence, and find which term it is.
- (e) Determine whether 300 is a term of the sequence. Justify your answer.

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2. In an arithmetic sequence, $u_5 = 31$ and $u_{12} = 3$.

(a) Find the common difference d and the first term u_1 .

(b) Find an expression for u_n .

(c) Find the first term of the sequence that is negative.

The first three terms of a different arithmetic sequence are

$$k + 2, \quad 3k, \quad 4k + 1.$$

(d) Find the value of k , and write down the first three terms.

(e) Find the 20th term of this sequence.

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3. (a) State whether each sequence is arithmetic, geometric or neither. Write down d or r where there is one.

(i) 3, 7, 11, 15 (ii) 3, 6, 12, 24 (iii) 3, 4, 6, 9 (iv) 80, 20, 5, 1.25

Consider the geometric sequence

6, 9, 13.5, ...

- (b) Find the common ratio r .
- (c) Write down an expression for u_n .
- (d) Find u_{10} .
- (e) Find the smallest value of n for which $u_n > 1000$.
- (f) A different geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find the common ratio and the first term.

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4. Elena is offered two jobs.

Job A: \$42 000 in the first year, rising by \$1500 each year.

Job B: \$40 000 in the first year, rising by 5% each year.

- (a) Find Elena's salary in year 3 for each job.
- (b) Write down an expression for the salary in year n for each job.
- (c) Find the first year in which the Job B salary is greater than the Job A salary.

Elena invests \$3000 at 3% per year compounded annually.

- (d) Find the value of the investment after 8 years.
- (e) Inflation is 2% per year. Find the real value of the investment after 8 years.

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